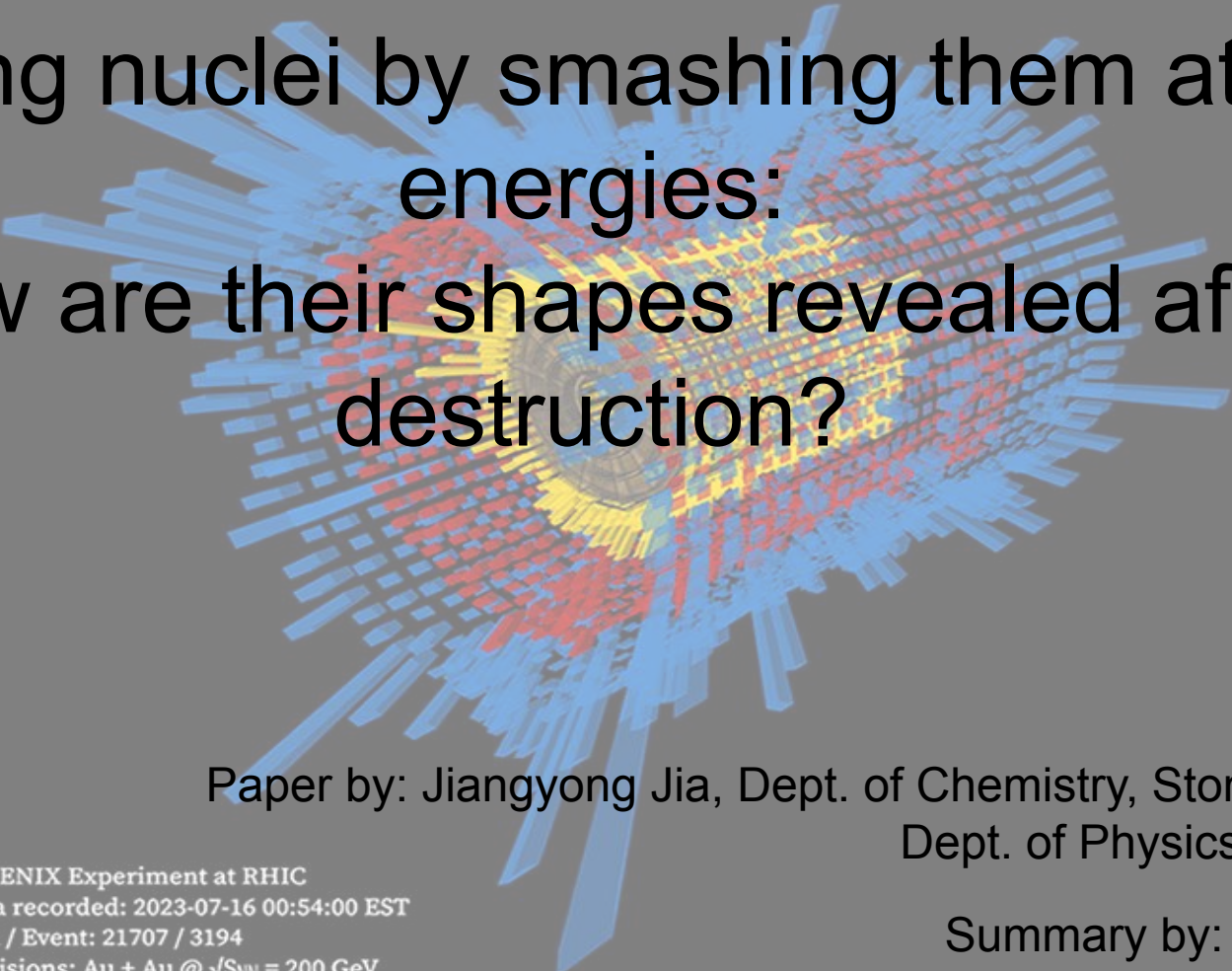


Imaging nuclei by smashing them at high energies: How are their shapes revealed after destruction?



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Summary by: Alec Ferensic



sPHENIX Experiment at RHIC
Data recorded: 2023-07-16 00:54:00 EST
Run / Event: 21707 / 3194
Collisions: Au + Au @ $\sqrt{s_{NN}} = 200$ GeV

What is Nuclear Imaging?



Nutshell: Using final state observables to probe the initial state anisotropy.

Photography Analogy:

- Negative = Momentum space correlations (from observables)
- Dark Room = Reverse engineer collision (how???)
- Photograph = Initial nuclear shape

Framework of discussion

Author's Questions:

“How are nuclear shapes encoded in such violent collisions?”

“How can they [nuclear shapes] be deduced from final-state particles?”

“What are the broader implications for nuclear physics?”

What understanding is needed to make an “image”

Initial conditions reconstructed using final-state observables and hydro

Can map initial conditions back to nuclear shape

Compare isobar collisions and their shapes

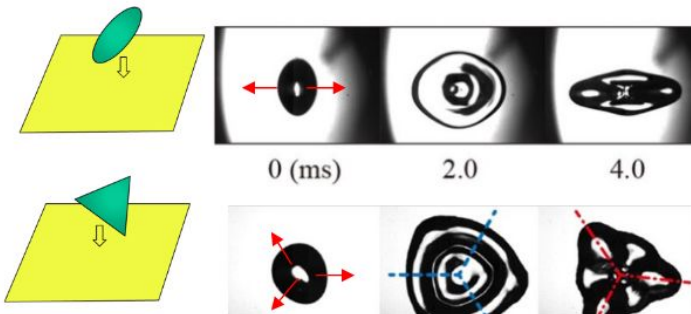
Traditional methods for nuclear shapes

(low energy)

Smashing a deformed water droplet on surface

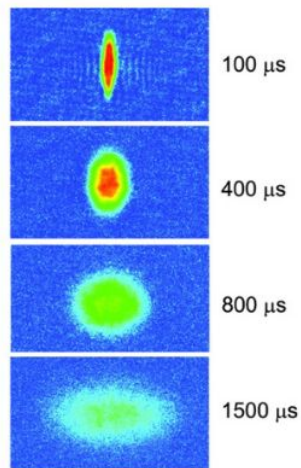
a

$$F = \nabla P$$



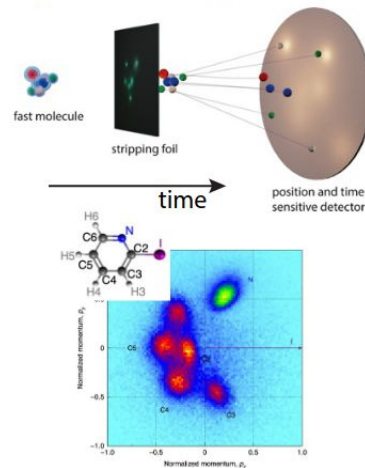
Expansion of strongly-coupled cold atomic gas

b

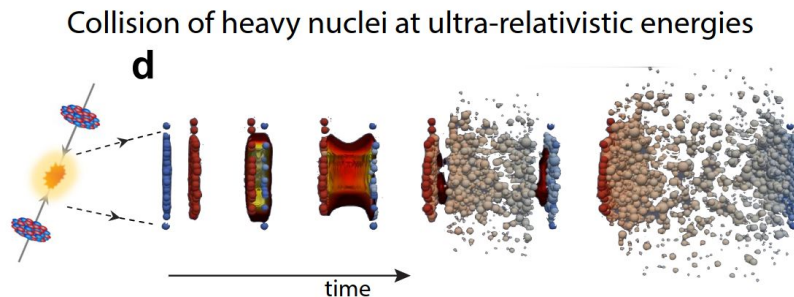


Coulomb Explosion Imaging in Chemistry

c



Imaging-by-smashing at high energy



- Collisions probe the GS nuclear many-body wave function in position space
- Expansion can be well described using hydrodynamics
-

Process is highly dependent on collision energy $\sqrt{s_{NN}}$

- Energy sets the time scale or “shutter speed”.
More energy = deeper probes

At 0.1 MeV
 $\tau \sim 1e3 - 1e4 \text{ fm/c}$
roughly $3e-20 \text{ sec}$

Reverse engineering nuclear shape

Nucleus quadropole deformation:

$$R(\theta, \phi) = R_0(1 + \beta_2(\cos \gamma Y_{2,0} + \sin \gamma Y_{2,2}))$$

Surface function with polar angle θ and azimuthal angle ϕ .

β is how non-spherical the nucleus is (“Quadropole Deformation”)

γ “controls the ratio of principal radii” prolate ($\gamma = 0^\circ$), oblate ($\gamma = 60^\circ$)

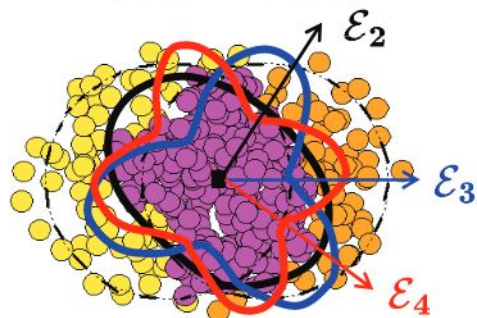
Components projected onto the xy-plane show initial geometries.

Jia, [Phys. Rev. C 105, 014905 \(2022\)](#), [arXiv:2106.08768 \[nucl-th\]](#).

Step 1: From the final state to the initial condition

a initial-state geometry

$$T(x, y) \text{ or } T(r_{\perp}, \phi)$$



$$\mathcal{E}_n \equiv \varepsilon_n e^{in\Phi_n} = -\frac{[r_{\perp}^n e^{in\phi}]}{[r_{\perp}^n;]}, \quad d_{\perp} = \frac{1}{[r_{\perp}^2]}$$

estimators

pressure-driven expansion of
quark-gluon plasma (QGP)

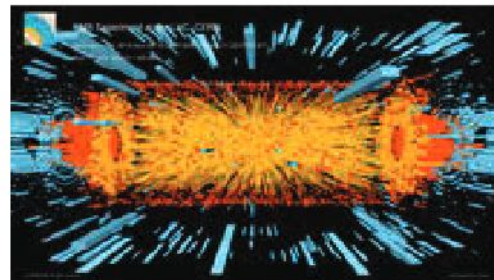
$$V_n \propto \mathcal{E}_n$$

$$\frac{\delta[p_T]}{[p_T]} \propto \frac{\delta d_{\perp}}{d_{\perp}}$$

linear response relations

b final-state particle flow

$$\frac{d^2N}{d\phi dp_T} = \frac{dN}{dp_T} \left[1 + 2 \sum_n v_n \cos(n(\phi - \Psi_n)) \right]$$



$$V_n = v_n e^{in\Psi_n}, [p_T] = \frac{\int p_T \frac{dN}{dp_T} dp_T}{\int \frac{dN}{dp_T} dp_T}$$

observables

$$d_{\perp} = 1/\sqrt{[x^2][y^2]},$$

$$\mathcal{E}_n \equiv \varepsilon_n e^{in\Phi_n} = -[(x + iy)^n] / [| (x + iy) |^n]$$

V is the flow coefficient
Measures azimuthal anisotropy

Step 1: From the final state to the initial condition

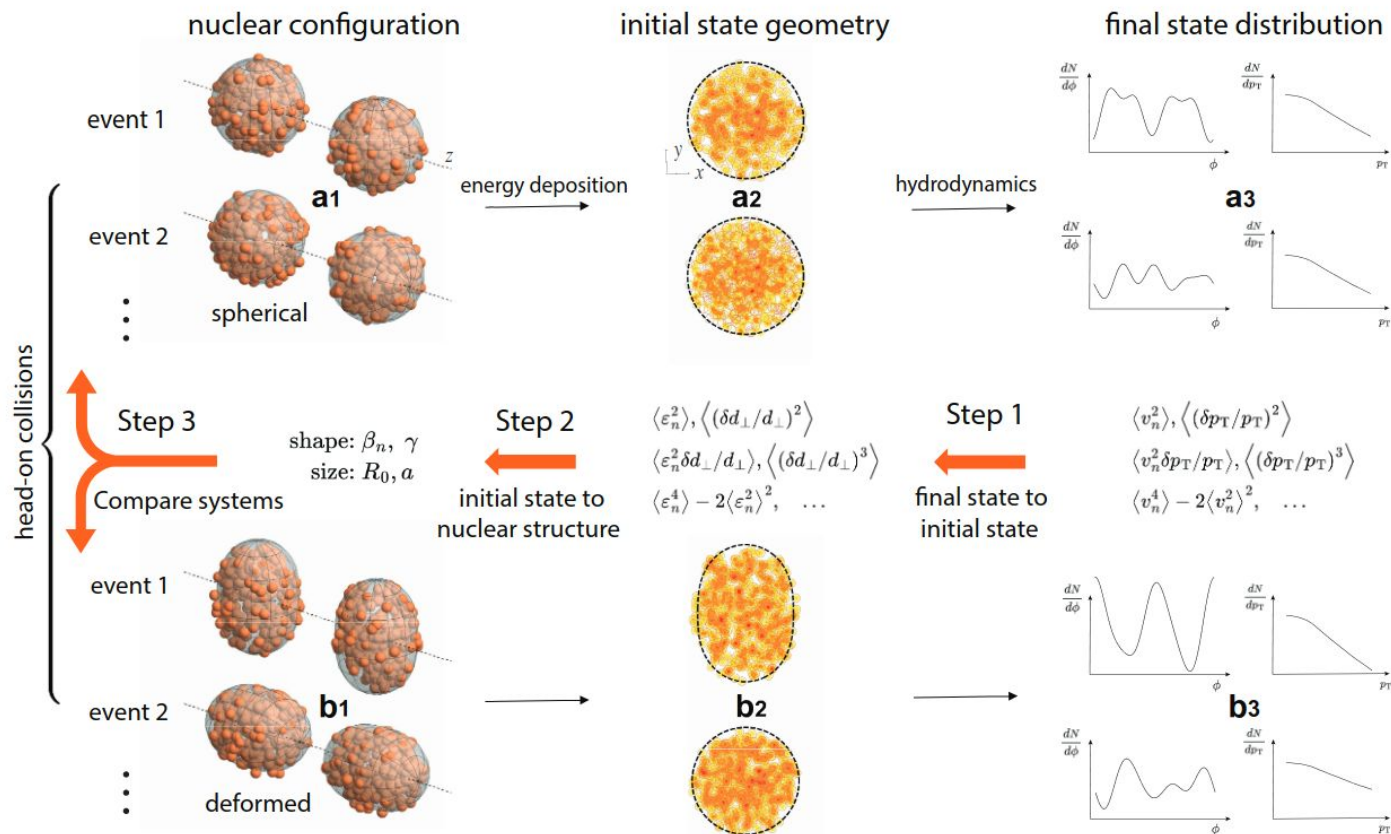
Goal: Get QGP initial geometry from fluctuations in observables of detected particles

Observables: Final state particles □ Momentum distributions, $\langle v_2^2 \rangle$, $\langle (\delta p_T)^2 \rangle$, and $\langle v_2^2 \delta p_T \rangle$. See STAR experiment

Linear Relations: Hydrodynamic expansion maps ε □ v approximately linearly

$$V_n \propto \mathcal{E}_n$$
$$\frac{\delta [p_T]}{[p_T]} \propto \frac{\delta d_\perp}{d_\perp}$$

Step 2: From the initial conditions to nuclear shape



Step 2: From the initial conditions to nuclear shape

Goal: Link momentum space fluctuations to nuclear deformation by showing how orientation changes the initial QGP geometry BEFORE hydrodynamic evolution.

- β describes how (un)spherical the object is while γ measures the triaxality.
 - Changes thickness function (Trento model):

$$T(x, y) = \left(\frac{T_A^p + T_B^p}{2} \right)^{q/p}$$

Step 2: From the initial conditions to nuclear shape

Observables:

$$\begin{aligned}\langle v_2^2 \rangle &= a_1 + b_1 \beta_2^2, & \langle \varepsilon_2^2 \rangle &= a'_1 + b'_1 \beta_2^2, \\ \langle (\delta p_T)^2 \rangle &= a_2 + b_2 \beta_2^2, & \langle (\delta d_\perp)^2 \rangle &= a'_2 + b'_2 \beta_2^2, \\ \langle v_2^2 \delta p_T \rangle &= a_3 - b_3 \beta_2^3 \cos(3\gamma) & \langle \varepsilon_2^2 \delta d_\perp \rangle &= a'_3 - b'_3 \beta_2^3 \cos(3\gamma)\end{aligned}$$

- Term a_n describes quantum fluctuation contributions.
- b_n reflects global deformation affects.
- $\cos(3\gamma)$ comes from triaxial rotational symmetry

Step 3: Separating global deformation from quantum fluctuations

Goal: Extract β , but a_n and b_n are modified by final-state effects.

“A more robust strategy involves comparing two collision systems of similar mass but different shapes”

Ratios of observables between a deformed nucleus and a spherical one:

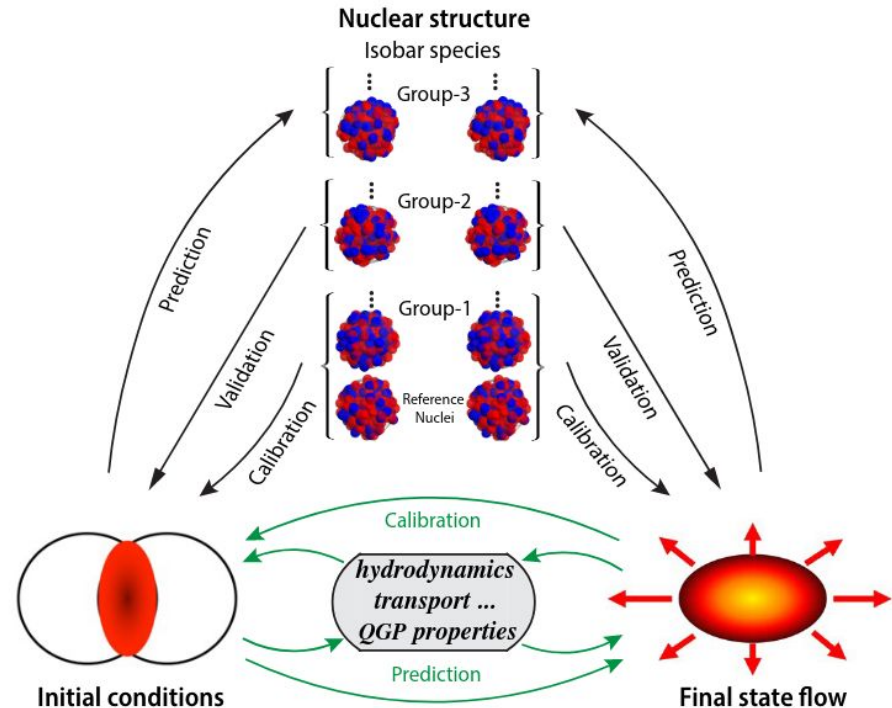
$$\begin{aligned}R_{\langle v_2^2 \rangle} &= 1 + \frac{b_1}{a_1} \beta_2^2 , \\ R_{\langle (\delta p_T)^2 \rangle} &= 1 + \frac{b_2}{a_2} \beta_2^2 , \\ R_{\langle v_2^2 \delta p_T \rangle} &= 1 - \frac{b_3}{a_3} \beta_2^3 \cos(3\gamma)\end{aligned}$$

Implementation of the Method

Calibration: Tuning response coefficients (RC's) using known nuclear structures. Typically done with a spherical nuclei and a highly deformed one for accuracy.

Validation: Check RC's against known shapes, expose any potential problems

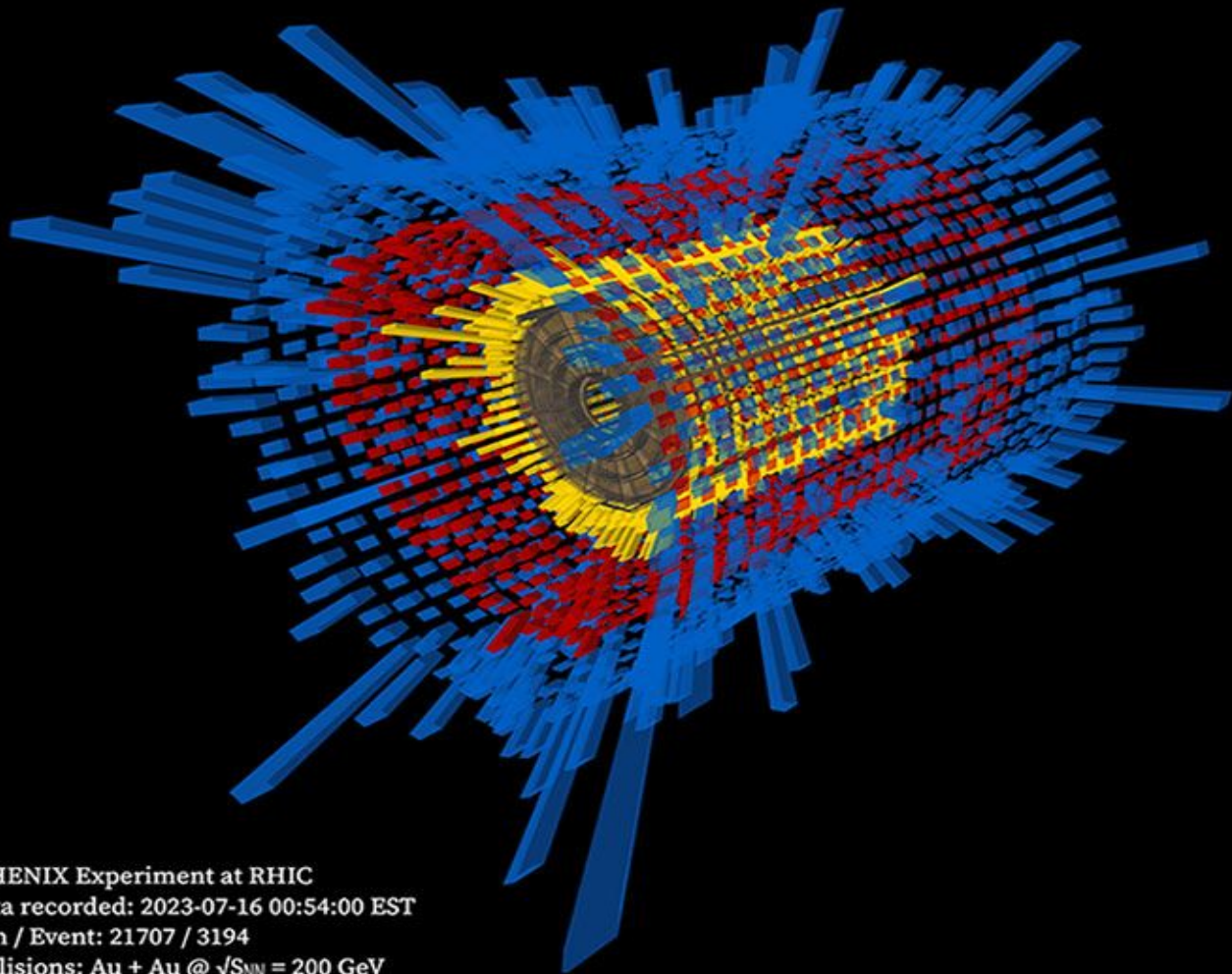
Prediction: Only once RC's are calibrated and validated, we can make possible discoveries.



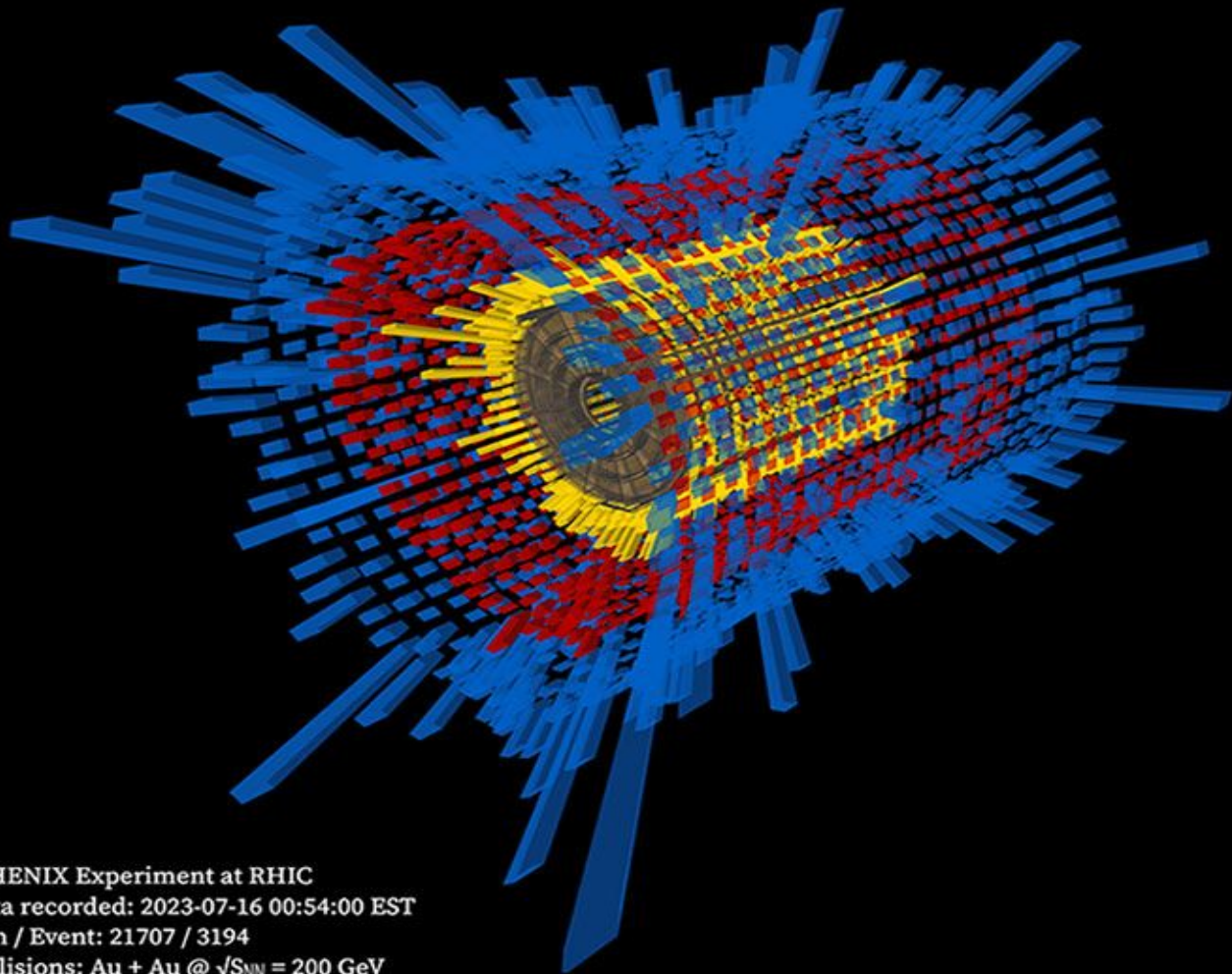
Conclusion

Final State -> Initial QGP Geometry -> Nuclear Shape

Event by event fluctuations in flow provide a probe for nuclear structure and is already demonstrated by STAR.



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